

Examiner reports

Q1.

In part (a), most correctly read the graph and then correctly used the coordinates to give the required P and V values.

The majority seemed well practised in the first step for part (b), writing the equation as $\log_{10}P = \log_{10}C + d\log_{10}V$. Many progressed to a correct solution easily and completed the question. Others became confused about what was a log and what was not or failed to handle the negative values and negative gradient. Those who tried to use point A and another point to generate simultaneous equations usually became lost at some point in the algebra.

In part (c), most remembered to include the units, relating the mathematical model back to the physical situation behind it.

Q3.

This question, which tested reducing a relation to a linear law, was generally answered well. In part (a) almost all students applied laws of logarithms, although some students failed to then clearly show the relationship was linear for Y and X . In part (b), students generally showed a good understanding of the methods required, although a minority mixed up Y and y in finding the value for a .

Q4.

Very few candidates scored full marks for this question.

Part (a) was well done with only a handful failing to recognise a stretch and most giving the correct direction and scale factor; most who were wrong in one of these were wrong in both. It was essential to use the correct term 'translate' for the other, so 'shift' did not earn marks and 'transformation' was inadequate. A few made an error in the vector but most were correct.

Most candidates were unsuccessful in answering part (b) of the question as their curve was either below the x -axis or continued into the second quadrant. Many placed the curve through e instead of $e + 1$, and the curvature, for $x < e + 1$, was often wrong.

Correct answers in part (c)(i) were sometimes spoiled by poor algebra. It was worrying to see the misunderstanding of the log function, with $\ln(x - e)$ expanded as $\ln x - \ln e$ and incorrect subsequent work once a correct answer had been found: $e^{-1} + e = 0$ was a common such response.

In part (c)(ii), although the first mark for $x \geq 2e$ was often earned (though $x \leq 2e$ was common), the second mark was only gained by a very few candidates, as the part of the inequality concerning e was generally omitted completely.

Q5.

Part (a) was generally correct. $\frac{e^0}{6}$ was not good enough and neither was 0.16. Wrong answers included 1 and $\ln 6$.

In part (b) many solutions were correct, often in an unsimplified form, but $\frac{1}{6x}$ and $\frac{6}{x}$ were common. Those who included dx or $+c$ in their solution were penalised.

Q6.

The candidates were mostly familiar with the techniques associated with logarithms, but the relationships between a and c , and between b and m , were often left in logarithmic form rather than in the way specified in the question, which involved exponential forms. These exponentials could perhaps have pointed candidates in the right direction when they came to answering part (c), where antilogarithms were needed for both answers. The linear graphs in part (b) were mostly plotted and drawn accurately, though some candidates misused the scale on the vertical axis.

Q7.

Part (a) was very familiar to the majority of candidates – perhaps **too** familiar in many cases as the result $\lg(ab^x) = \lg a + x \lg b$ was quoted rather than properly shown by the use of the laws of logarithms.

Part (b) presented a less familiar type of challenge. It was not necessary or indeed helpful to try to find values for the constants m and c . All that was needed was to read values directly from the graph and to convert as appropriate between y and Y .

Q8.

Part (a) of this question was well answered, most candidates being familiar with the equation $y = ax^b$ and the technique needed to convert it into linear form. Some candidates seemed less happy with part (b), but most managed to show enough knowledge to score well here. Errors often arose from confusion between the intercept 1 on the vertical axis and the corresponding value of y , which required the taking of an antilogarithm.

Q9.

The great majority of candidates gave the correct values to three decimal places in part (a). Most of them coped efficiently with the logarithmic manipulation in part (b), but occasionally a candidate would treat the expression ab^x as if it were $(ab)^x$, resulting in a loss of at least four marks, as it was now impossible to distinguish validly between a and b .

The plotting of the points on the graph was usually well done, but some candidates misread the vertical scale here, and again in part (d) when they attempted to read off the intercept on the Y – axis. Some candidates failed to use this intercept, resorting instead to more indirect methods of finding a value for $\log a$. Methods for finding the gradient of the linear graph were equally clumsy. Some candidates, even after obtaining a correct equation in part (b), did not realise that the taking of antilogs was needed in part (d).



Q10.

Very high marks were scored in this question. With few exceptions the values in the table and the points on the graph were accurately given, the work on logarithms was done well, and in part (d) the candidates showed an awareness that the gradient of the graph was needed. Sometimes an unfortunate choice of points was used as the basis for the calculation of this gradient.



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